

Smart Multi-Authentication Solenoid Door Lock System using ESP8266 NodeMCU

Embedded Systems and IoT Project Report

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Abstract

The Smart Multi-Authentication Solenoid Door Lock System using ESP8266 NodeMCU is an embedded security automation project designed to provide secure and flexible access control using multiple authentication techniques.

The project integrates RFID authentication, matrix keypad password verification, fingerprint-based biometric authentication, and serial monitor command control for operating a relay-controlled solenoid locking mechanism.

The system was implemented using real-time hardware interfacing and demonstrates practical applications of embedded systems, sensor interfacing, authentication systems, firmware development, and IoT concepts.

Introduction

Modern security systems require reliable and intelligent authentication mechanisms to prevent unauthorized access. Traditional locking systems often rely on a single authentication mechanism, making them less secure and less flexible.

This project focuses on developing a smart multi-authentication embedded door lock system capable of supporting multiple authentication techniques including RFID verification, keypad password authentication, fingerprint recognition, and serial monitor control.

The project was developed to strengthen practical understanding of embedded systems, IoT concepts, sensor interfacing, and real-time hardware integration through hands-on implementation.

Objectives

- Develop a smart multi-authentication embedded security system.
- Implement RFID-based authentication using RC522 module.
- Implement keypad password authentication.
- Implement fingerprint-based biometric authentication.
- Implement serial monitor command control.
- Interface relay and solenoid lock for door operation.
- Strengthen practical knowledge in embedded systems and IoT.

Components Used

ESP8266 NodeMCU	Main Controller
RC522 RFID Module	RFID Authentication
4x4 Matrix Keypad	Password Authentication
R307 Fingerprint Sensor	Biometric Authentication
Relay Module	Solenoid Lock Control
12V Solenoid Lock	Door Locking Mechanism
Buzzer	Access Feedback
Breadboard	Hardware Prototyping
Jumper Wires	Electrical Connections
External Power Supply	Solenoid Power Source

Real-Time Hardware Implementation

The project was implemented and tested using real-time hardware interfacing with ESP8266 NodeMCU and multiple authentication modules.

Each authentication system was individually developed, tested, and verified before integration.

RFID Authentication

The RFID authentication module uses RC522 RFID reader to verify authorized RFID cards using UID comparison logic.

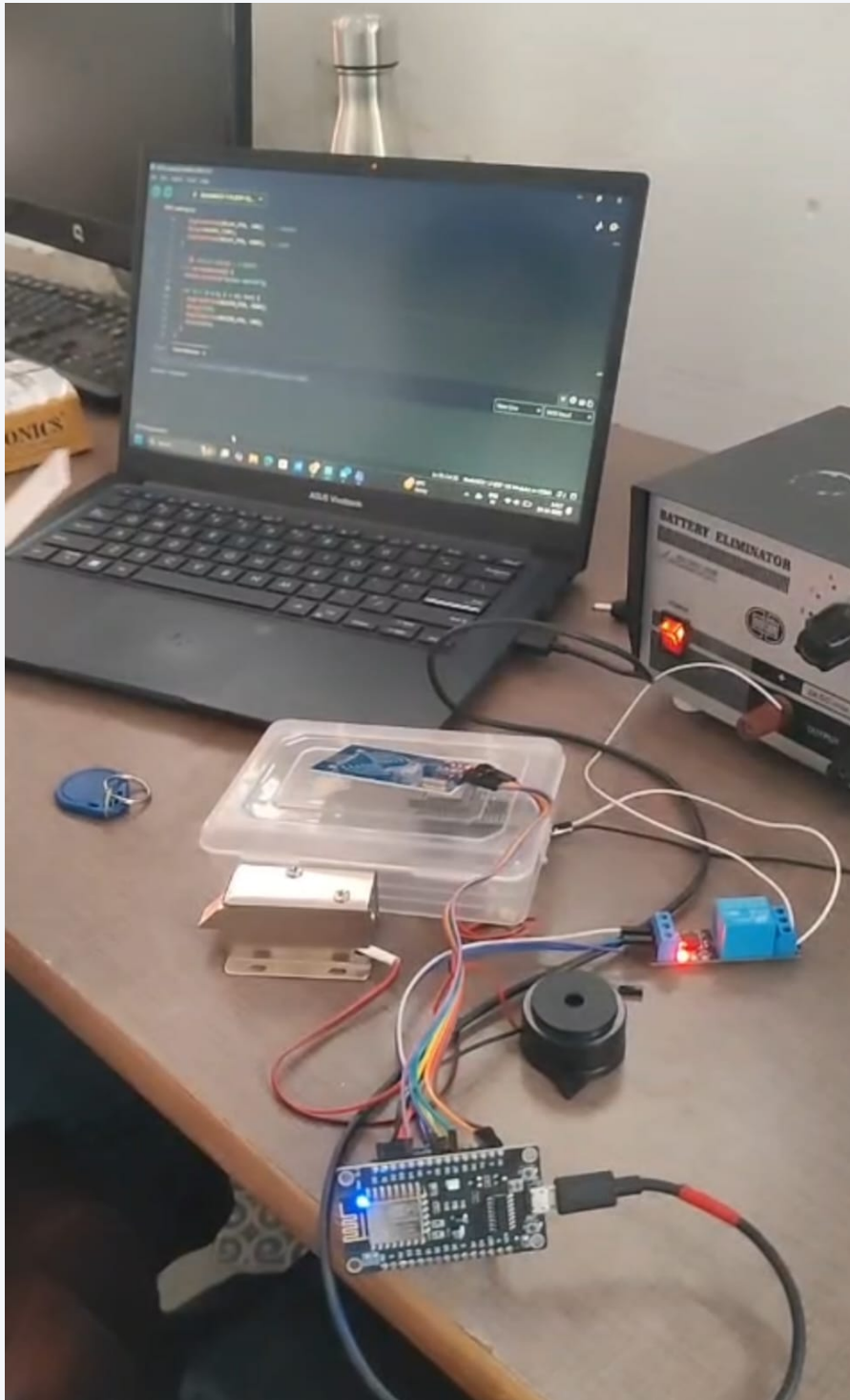


Figure 1: Real-Time RFID Authentication Implementation

Serial Monitor Command Control

The serial monitor authentication system controls the relay-operated solenoid lock using ON and OFF commands through Arduino IDE Serial Monitor.

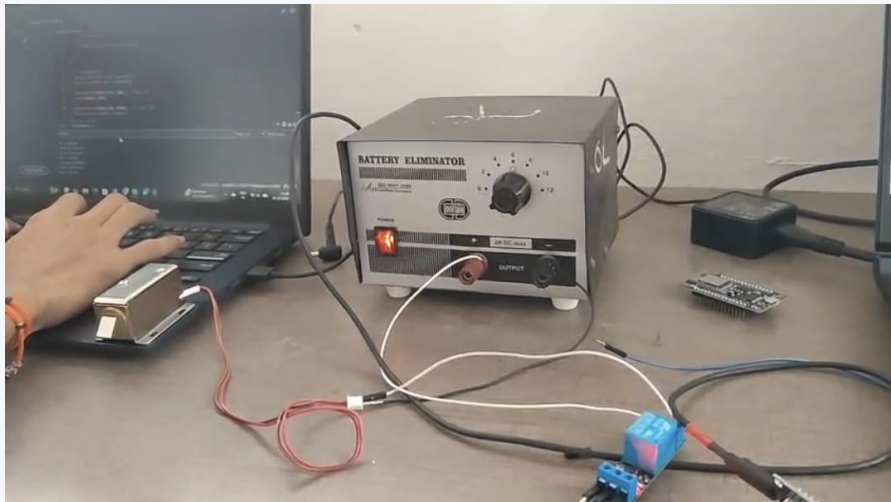


Figure 2: Real-Time Serial Monitor Control Implementation

Matrix Keypad Authentication

The keypad authentication module provides password-based access verification using a 4×4 matrix keypad connected to ESP8266 NodeMCU.

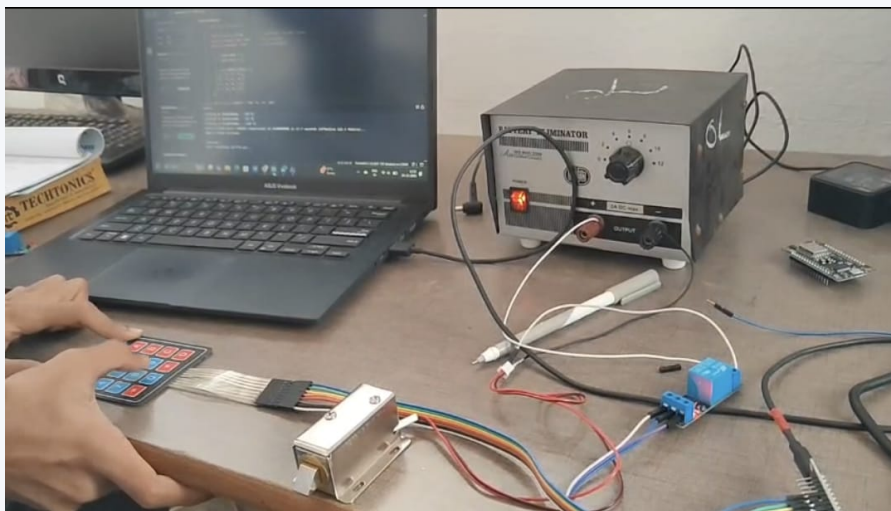


Figure 3: Real-Time Keypad Authentication Implementation

Fingerprint Authentication

The fingerprint authentication module uses R307 fingerprint sensor for biometric verification before unlocking the solenoid lock.



Figure 4: Real-Time Fingerprint Authentication Implementation

Firmware Code Snippets

RFID Authentication Logic

```
if (isAuthorized()) {  
    unlockDoor();  
}  
else {  
    accessDenied();  
}
```

Keypad Password Verification

```
if (input == PASSWORD) {  
    unlockDoor();  
}
```

Fingerprint Verification

```
p = finger.fingerFastSearch();  
  
if (p == FINGERPRINT_OK) {  
    unlockDoor();  
}
```

Serial Monitor Command Control

```
if (command == "ON") {  
    unlockDoor();  
}
```

Testing and Validation

RFID Authentication	Passed
Unauthorized RFID Detection	Passed
Keypad Password Authentication	Passed
Fingerprint Authentication	Passed
Serial ON/OFF Control	Passed
Relay-Based Solenoid Locking	Passed

Applications

- Smart Home Security Systems
- Office Access Control
- Embedded Security Systems
- IoT-Based Door Automation
- Biometric Authentication Systems
- Industrial Access Management

Advantages

- Multiple authentication methods
- Enhanced security architecture
- Real-time embedded implementation
- Product-oriented design
- Modular firmware structure
- Low-cost embedded solution

Future Scope

Future enhancements may include:

- ESP32-CAM QR code authentication
- Mobile application integration
- Face recognition authentication
- Cloud-based monitoring system
- Firebase database integration
- PCB-based compact design

Project Repository

GitHub Repository:

[multi-authentication-door-lock-system](#)



Figure 5: GitHub Repository QR Code

Conclusion

The Smart Multi-Authentication Solenoid Door Lock System using ESP8266 NodeMCU successfully demonstrated secure embedded access control using RFID authentication, keypad password verification, fingerprint authentication, and serial monitor command control.

The project strengthened practical understanding of embedded systems, firmware development, sensor interfacing, hardware integration, and IoT concepts through real-time implementation and testing.

References

- ESP8266 NodeMCU Documentation
- RC522 RFID Module Datasheet
- Adafruit Fingerprint Sensor Library
- Arduino IDE Documentation
- Embedded Systems and IoT Reference Materials